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Report

Laboratory Work No. 2

Cryptography and Security

Topic: Cryptanalysis of Monoalphabetic Ciphers

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1 INTRODUCTION

1.1 Monoalphabetic ciphers and their weak point

A monoalphabetic cipher replaces every letter of the plaintext with another letter, always using the same substitution alphabet. The Caesar cipher from Laboratory Work No. 1 is the simplest example; in the general case the substitution alphabet can be any permutation of the 26 letters, so there are $26! \approx 4 \times 10^{26}$ possible keys and an exhaustive search is impossible.

The weak point of all monoalphabetic ciphers is that they do not change the statistics of the language. A given plaintext letter is always encrypted into the same ciphertext letter, so the most frequent letter of the plaintext language (for English, E) becomes the most frequent symbol of the cryptogram, the most common words such as THE, AND, THAT keep their shape, and so on. If the ciphertext is long enough and the language of the message is known, the cipher can be broken by frequency analysis.

1.2 Purpose of the laboratory work

The purpose of this laboratory work is to study frequency analysis, the classical attack on monoalphabetic ciphers, and to apply it in practice to recover an English text from an intercepted cryptogram.

Task. An encrypted message has been intercepted. It is known that it was obtained with a monoalphabetic cipher and that the plaintext is written in English. Only the letters were encrypted; all other characters (spaces, punctuation, digits) were left unchanged. Using frequency analysis, recover the original message and describe the breaking process in the same way as in the example from Section 2.3 of the laboratory guide. The assigned variant is **V4**.

To compute the letter frequencies, the online tool recommended in the task was used: Crypto Corner "Frequency Analysis: Breaking the Code" (crypto.interactive-maths.com).

2 THEORETICAL BACKGROUND

2.1 Frequency analysis

Every natural language uses its letters with a characteristic frequency. In a long enough English text, E accounts for about 12.7% of all letters, T for about 9.1%, A for 8.2%, while J, X, Q and Z appear in less than 0.2% of cases (see the upper row of Figure 3.1). The longer the ciphertext, the closer the frequencies of its symbols come to this general distribution. Comparing the order of the ciphertext symbols with the order of the letters of the language gives the first, most probable correspondences between ciphertext and plaintext letters.

2.2 Methodology of the attack

Replacing all letters strictly according to the frequency order almost never gives the correct text, because the frequencies of a particular message differ slightly from the average ones. Frequency analysis is therefore combined with the structure of the language:

The most frequent symbol is first assumed to be E, and the next ones T, A, O, I, N. Then the most common English words are used as clues: one-letter words can only be A or I, the most common three-letter word is THE, and two-letter words such as OF, TO, IN, IS, IT are very frequent. After each substitution, partially decrypted words are completed to the only English words that fit, which reveals new letters, until the whole text can be read.

In the partially decrypted texts below, the letters that are still encrypted are written in UPPERCASE and the letters already found are written in lowercase, exactly as in the example from the laboratory guide.

3 BREAKING THE CRYPTOGRAM

3.1 The intercepted cryptogram

The intercepted cryptogram for variant V4 is (the capital letters of the original are kept in this listing; in the analysis below the case is ignored):

```
Odixgj tss wqvpv fvtip, hifuwnsnjf rtp thbdxixgj t wtxgw wqtw sxgjvipvkvg wnotf—wqv
hngkxhwxng xg wqv zxgop nc ztgf uvnusv wqtw hifuwnsnjfxp t asthl tiw, t cniz nc nhhdswxpz rqnpv
uithwxwxngvi zdpw, xg Rxssxtz C.Cixvoztg'p tuw uqitpv, "uvicnihv hnzzdgv otxsf rxwq otil puxixwp
wnthhnzusxpq qxp cvtwp nc zvgwts exd-exwpd."Xg utiw xw xp t lxgo nc jdxsw af tppnhxtwxng.
Cinz wqv vtisf otfp nc xwpvyxpwvghv, hifuwnsnjf qto pvikvo wn naphdiv hixwxhts uniwxngp nc
rixwxgjpovtsxgj rxwq wqv unwvgw pdaevhw nc ztjxh—oxkxgtwxngp, puvssp, hdipvp,rqtwvkvi
hngcviivo pduvigtwdits unrvip ng xwp pnihvivip. Tgnwqvixzuniwtgw cthwni rtp wqv hngcdpxng nc
hifuwnsnjf rxwq wqv Evrxpqltaatstq.Adw, xzuniwtgw tp tss wqvpv rviv, wqv kxvr wqtw hifuwnsnjf
xp asthlztjxh xg xwpvsc puixgjp dswxztwvsf cinz t pduvicxhxts ivpvzastghv avwrvvghifuwnsnjf tgo
oxkxgtwxng. Vywithwxgj tg xgwvssxjxasv zvpptjv cinzhxuqviwvyw pvvzvo wn av vythwsf wqv ptzv
wqxgj tp nawtxgxgj lgnrsvojvaf vytzxgxgj wqv csxjqw nc axiop, wqv snhtwxng nc pwtip tgo
ustgvwp, wqvsvgjwq tgo xgwvipvhwxngp nc sxgvp xg wqv qtgo, wqv vgwitxsp nc pqvvu,
wqvunpxwxng nc oivjp xg t wvthdu. Xg tss nc wqvpv, wqv rxmtio-sxlv nuvitwnioitrp pvgpv cinz
jinwvpbdv, dgctzxsxti, tgo tuutivgwsf zvtgxgjsvpppxjgp. Qv ztlvp lgnrg wqv dglgnrg.Tss wqxp
pwtxgvo hifuwnsnjf pn ovvusf rxwq wqv otil qdvp nc vpnwvixpzwqtw pnzv nc wqvz pwxss uvipxpw,
gnwxhvtasf hnsnixgj wqv udasxh xztjv nchifuwnsnjf. Uvnusv pwxss wqxgl hifuwtgtsfpxp
zfpwvixndp. Annl ovtsvip pwxsssxpw hifuwnsnjf dgovi "nhhdsw." Tgo xg 1940 wqv Dgxwvo
Pwtwvp hngcviivodung xwp Etutgvpv oxusnztwxh hifuwtgtsfpvp wqv hnovgtzv ZTJXH. Xg gngv nc
wqv pvhivw rixwxgj wqdp cti rtp wqviv tgf pdpwtxgvohifuwtgtsfpxp. Nhhtpxngts htpvp, fvp. Adw
nc tgf phxvghv nc hifuwtgtsfpxp,wqviv rtp gnwqxgj. Ngsf hifuwnjituqf vyxpwvo. Tgo wqvivcniv
hifuwnsnjf,rqxhq xgknskvp anwq hifuwnjituqf tgo hifuwtgtsfpxp, qto gnw fvw hnzv xgwn avxgj pncti
tp tss wqvpv hdswdivp—xghsdoxgj wqv Rvpwvig —rviv hnghvigvo.Hifuwnsnjf rtp anig tzngj wqv
Titap. Wqvf rviv wqv cxipw wn oxphnkvitgo rixwv onrg wqv zvwqnop nc hifuwtgtsfpxp. Wqv
uvnusv wqtw vyusnovondw nc Titaxt xg wqv 600p tgo cstzvo nkvi ktpw tivtp nc wqv lgnrg
rnisoprxcwsf vgjvgovivo ngv nc wqv qxjqvpw hxkxsxmtwxngp wqtw qxpwnif -qto fvwpvvg.
Phxvghv csnrvivo. Tita zvoxhxgv tgo ztwqvztwxhp avhtzv wqv avpwxg wqv rniso—cinz wqv
stwwvi, xg cthw, hnzvp wqv rnio "hxuqvi." Uithwxhtstiwp csndixpqvo. Tozxgxpwitwxkv
wvhqgxbdvp ovkvsnuvo. Wqv vydavitgwhivtwxkv vgvijxvp nc pdhq t hdswdiv, vyhsdovo af xwp
ivsxjxng cinz utxgwxgjni phdsuwdiv, tgo xgpuxivo af xw wn tg vyusxhtwxng nc wqv Qnsf
Lnitg,undivo xgwn sxwvitif udipdxwp. Pwnifwvssxgj, vyvzusxcxvo af Pqvqvitmtov'pWqndptgo tgo
Ngv Gxjqwp, rnio-ixoosvp, ivadpvp, udgp, tgtjitzp, tgopxzxsti jtzvp tandgovo; jitzzti avhtzv t zteni
pwdof. Tgo xghsdovortp pvhivw rixwxgj.Tcwvi vyustxgxgj wqtw ngv ztf rixwv xg tg dglgnrg stgjdtjv
wn nawtxgpvhivhf, Xag to-Oditxqxz, thhnioxgj wn Btsbtpqtgox, jtkv pvkvg pfpwvzpnc hxuqvip.
Wqxp sxpw vghnzutppvo, cni wqv cxipw wxzv xg hifuwnjituqf, anwqwitgpunpxwxng tgo
pdapwxwdwxng hxuqvip. Znivnkvi, ngv pfpwvz xp wqv cxipwlgnrg hxuqvi vkvi wn uinkxov zniv
wqtg ngv pdapwxwdwv cni t ustxgwvywsvwwvi. Ivztiltasv tgo xzuniwtgw tp wqxp xp, qnrvkvi, xw
xp nkvipqtonrvo af rqtw cnssnrp— wqv cxipwvyunpxwxng ng hifuwtgtsfpxp xg qxpwnif.
```

3.2 Letter frequencies of the cryptogram

The first step is to find the frequencies of all letters in the cryptogram. The text contains 2754 letters; all 26 letters of the alphabet occur in it. Figure 3.1 shows the result given by the Crypto Corner tool, with the letters sorted in decreasing order of frequency.

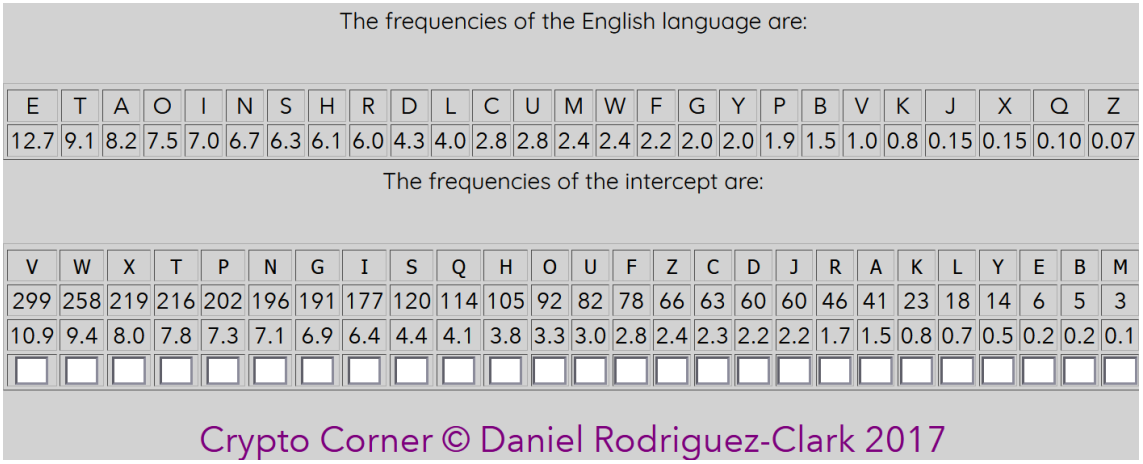


Figure 3.1 – Frequencies of the English language and of the intercept (Crypto Corner)

The distribution of the cryptogram has the same shape as that of English: one clear leader (V, 10.9%), a group of letters between 6.4% and 9.4%, and a long tail of rare letters. This confirms that the letters were only renamed, as in a monoalphabetic cipher.

3.3 Step-by-step substitutions

Each step below gives the reasoning, the substitution that was made, and the first 8 lines of the text after the substitution (the complete decrypted text is given in Section 3.4).

Step 1 – the most frequent letter. The most frequent letter of the cryptogram is „V" (299 occurrences, 10.9%), clearly ahead of „W" (9.4%). In English the most frequent letter is „e" (12.7%), so the first assumption is V → e.

Substitution: „V" → „e"

```
ODIXGJ TSS WQePe FeTIP, HIFUWNSNJF RTP THBDXIXGJ T WTXGW WQTW SXGJeIPeKeG WNOTF—WQe
HNGKXHWXNG XG WQe ZXGOP NC ZTGF UeNUSe WQTW HIFUWNSNJFXP T ASTHL TIW, T CNIZ NC NHHDSWXPZ RQNPe
UITHWXWXNGeI ZDPW, XG RXSSXTZ C.CIXeOZTG'P TUW UQITPe, "UeICNIHe HNZZDGe OTXSF RXWQ OTIL PUXIXWP
WNTHHNZUSXPQ QXP CeTWP NC ZeGWTS EXD-EXWPD."XG UTIW XW XP T LXGO NC JDXSW AF TPPNHXTWXNG.
CINZ WQe eTISF OTFP NC XWPeYXPWeGHe, HIFUWNSNJF QTO PeIKeO WN NAPHDIe HIXWXHTS UNIWXNGP NC
RIXWXGJPOeTSXGJ RXWQ WQe UNWeGW PDAEeHW NC ZTJXH—OXKXGTWXNGP, PUeSSP, HDIPeP,RQTWeKeI
HNGCeIIeO PDUeIGTWDITS UNReIP NG XWP PNIHeIeIP. TGNWQeIXZUNIWTGW CTHWNI RTP WQe HNGCDPXNG NC
HIFUWNSNJF RXWQ WQe EeRXPQLTAATSTQ.ADW, XZUNIWTGW TP TSS WQePe ReIe, WQe KXeR WQTW HIFUWNSNJF
```

Step 2 – the word "the". After the first substitution the three-letter group „WQe" appears very often in the text (for example „WQe ZXGOP NC ZTGF" in the second line). The most common three-letter word in English is "the", so W → t and Q → h. This agrees with the frequencies: „W" is the second most frequent letter of the cryptogram (9.4%) and „t" is the second most frequent letter of English (9.1%).

Substitution: „W" → „t", „Q" → „h"

```
ODIXGJ TSS thePe FeTIP, HIFUtNSNJF RTP THBDXIXGJ T tTXGt thTt SXGJeIPeKeG tNOTF—the
HNGKXHtXNG XG the ZXGOP NC ZTGF UeNUSe thTt HIFUtNSNJFXP T ASTHL TIt, T CNIZ NC NHHDStXPZ RhNPe
UITHtXtXNGeI ZDPt, XG RXSSXTZ C.CIXeOZTG'P TUt UhITPe, "UeICNIHe HNZZDGe OTXSF RXth OTIL PUXIXtP
tNTHHNZUSXPh hXP CeTtP NC ZeGtTS EXD-EXtPD."XG UTIt Xt XP T LXGO NC JDXSt AF TPPNHXTtXNG.
CINZ the eTISF OTFP NC XtPeYXPteGHe, HIFUtNSNJF hTO PeIKeO tN NAPHDIe HIXtXHTS UNItXNGP NC
RIXtXGJPOeTSXGJ RXth the UNteGt PDAEeHt NC ZTJXH—OXKXGTtXNGP, PUeSSP, HDIPeP,RhTteKeI
HNGCeIIeO PDUeIGTtDITS UNReIP NG XtP PNIHeIeIP. TGNtheIXZUNItTGt CTHtNI RTP the HNGCDPXNG NC
HIFUtNSNJF RXth the EeRXPhLTAATSTh.ADt, XZUNItTGt TP TSS thePe ReIe, the KXeR thTt HIFUtNSNJF
```

Step 3 – the word "that". The word „thTt" now appears several times (e.g. „the ZXGOP NC ZTGF UeNUSe thTt…"). The only English word of this form is "that", so T → a. „T" is the 4th most frequent letter of the cryptogram (7.8%) and „a" the 3rd of English (8.2%).

Substitution: „T" → „a"

```
ODIXGJ aSS thePe FeaIP, HIFUtNSNJF RaP aHBDXIXGJ a taXGt that SXGJeIPeKeG tNOaF—the
HNGKXHtXNG XG the ZXGOP NC ZaGF UeNUSe that HIFUtNSNJFXP a ASaHL aIt, a CNIZ NC NHHDStXPZ RhNPe
UIaHtXtXNGeI ZDPt, XG RXSSXaZ C.CIXeOZaG'P aUt UhIaPe, "UeICNIHe HNZZDGe OaXSF RXth OaIL PUXIXtP
tNaHHNZUSXPh hXP CeatP NC ZeGtaS EXD-EXtPD."XG UaIt Xt XP a LXGO NC JDXSt AF aPPNHXatXNG.
CINZ the eaISF OaFP NC XtPeYXPteGHe, HIFUtNSNJF haO PeIKeO tN NAPHDIe HIXtXHaS UNItXNGP NC
RIXtXGJPOeaSXGJ RXth the UNteGt PDAEeHt NC ZaJXH—OXKXGatXNGP, PUeSSP, HDIPeP,RhateKeI
HNGCeIIeO PDUeIGatDIaS UNReIP NG XtP PNIHeIeIP. aGNtheIXZUNItaGt CaHtNI RaP the HNGCDPXNG NC
HIFUtNSNJF RXth the EeRXPhLaAAaSah.ADt, XZUNItaGt aP aSS thePe ReIe, the KXeR that HIFUtNSNJF
```

Step 4 – the word "than" / "that". Next we notice the word „thaG". It can only be "that" or "than"; since „t" has already been assigned to „W", „G" must be „n".

Substitution: „G" → „n"

```
ODIXnJ aSS thePe FeaIP, HIFUtNSNJF RaP aHBDXIXnJ a taXnt that SXnJeIPeKen tNOaF—the
HNnKXHtXNn Xn the ZXnOP NC ZanF UeNUSe that HIFUtNSNJFXP a ASaHL aIt, a CNIZ NC NHHDStXPZ RhNPe
UIaHtXtXNneI ZDPt, Xn RXSSXaZ C.CIXeOZan'P aUt UhIaPe, "UeICNIHe HNZZDne OaXSF RXth OaIL PUXIXtP
tNaHHNZUSXPh hXP CeatP NC ZentaS EXD-EXtPD."Xn UaIt Xt XP a LXnO NC JDXSt AF aPPNHXatXNn.
CINZ the eaISF OaFP NC XtPeYXPtenHe, HIFUtNSNJF haO PeIKeO tN NAPHDIe HIXtXHaS UNItXNnP NC
RIXtXnJPOeaSXnJ RXth the UNtent PDAEeHt NC ZaJXH—OXKXnatXNnP, PUeSSP, HDIPeP,RhateKeI
HNnCeIIeO PDUeInatDIaS UNReIP Nn XtP PNIHeIeIP. anNtheIXZUNItant CaHtNI RaP the HNnCDPXNn NC
HIFUtNSNJF RXth the EeRXPhLaAAaSah.ADt, XZUNItant aP aSS thePe ReIe, the KXeR that HIFUtNSNJF
```

Step 5 – the words "as" and "these". The text contains the two-letter word „aP" and the five-letter word „thePe" (in „aSS thePe FeaIP"). „thePe" can be "there" or "these", but „aP" cannot be "ar", so P → s, giving "as" and "these".

Substitution: „P" → „s"

```
ODIXnJ aSS these FeaIs, HIFUtNSNJF Ras aHBDXIXnJ a taXnt that SXnJeIseKen tNOaF—the
HNnKXHtXNn Xn the ZXnOs NC ZanF UeNUSe that HIFUtNSNJFXs a ASaHL aIt, a CNIZ NC NHHDStXsZ RhNse
UIaHtXtXNneI ZDst, Xn RXSSXaZ C.CIXeOZan's aUt UhIase, "UeICNIHe HNZZDne OaXSF RXth OaIL sUXIXts
tNaHHNZUSXsh hXs Ceats NC ZentaS EXD-EXtsD."Xn UaIt Xt Xs a LXnO NC JDXSt AF assNHXatXNn.
CINZ the eaISF OaFs NC XtseYXstenHe, HIFUtNSNJF haO seIKeO tN NAsHDIe HIXtXHaS UNItXNns NC
RIXtXnJsOeaSXnJ RXth the UNtent sDAEeHt NC ZaJXH—OXKXnatXNns, sUeSSs, HDIses,RhateKeI
HNnCeIIeO sDUeInatDIaS UNReIs Nn Xts sNIHeIeIs. anNtheIXZUNItant CaHtNI Ras the HNnCDsXNn NC
HIFUtNSNJF RXth the EeRXshLaAAaSah.ADt, XZUNItant as aSS these ReIe, the KXeR that HIFUtNSNJF
```

Step 6 – the words "is", "it", "in". Very short words such as „Xs", „Xt" and „Xn" are now frequent („Xt Xs a LXnO…"). Since „a" and „e" are already taken, the only English letter that makes all of them words is „i": "is", "it", "in". So X → i, which also fits the frequency of „X" (8.0%, the third most frequent).

Substitution: „X" → „i"

```
ODIinJ aSS these FeaIs, HIFUtNSNJF Ras aHBDiIinJ a taint that SinJeIseKen tNOaF—the
HNnKiHtiNn in the ZinOs NC ZanF UeNUSe that HIFUtNSNJFis a ASaHL aIt, a CNIZ NC NHHDStisZ RhNse
UIaHtitiNneI ZDst, in RiSSiaZ C.CIieOZan's aUt UhIase, "UeICNIHe HNZZDne OaiSF Rith OaIL sUiIits
tNaHHNZUSish his Ceats NC ZentaS EiD-EitsD."in UaIt it is a LinO NC JDiSt AF assNHiatiNn.
CINZ the eaISF OaFs NC itseYistenHe, HIFUtNSNJF haO seIKeO tN NAsHDIe HIitiHaS UNItiNns NC
RIitinJsOeaSinJ Rith the UNtent sDAEeHt NC ZaJiH—OiKinatiNns, sUeSSs, HDIses,RhateKeI
HNnCeIIeO sDUeInatDIaS UNReIs Nn its sNIHeIeIs. anNtheIiZUNItant CaHtNI Ras the HNnCDsiNn NC
HIFUtNSNJF Rith the EeRishLaAAaSah.ADt, iZUNItant as aSS these ReIe, the KieR that HIFUtNSNJF
```

Step 7 – the word "none". The word „nNne" („in nNne NC the seHIet…") can only be "none", since „i" is already taken. So N → o. The frequency of „N" (7.1%) is very close to that of „o" in English (7.5%).

Substitution: „N" → „o"

```
ODIinJ aSS these FeaIs, HIFUtoSoJF Ras aHBDiIinJ a taint that SinJeIseKen toOaF—the
HonKiHtion in the ZinOs oC ZanF UeoUSe that HIFUtoSoJFis a ASaHL aIt, a CoIZ oC oHHDStisZ Rhose
UIaHtitioneI ZDst, in RiSSiaZ C.CIieOZan's aUt UhIase, "UeICoIHe HoZZDne OaiSF Rith OaIL sUiIits
toaHHoZUSish his Ceats oC ZentaS EiD-EitsD."in UaIt it is a LinO oC JDiSt AF assoHiation.
CIoZ the eaISF OaFs oC itseYistenHe, HIFUtoSoJF haO seIKeO to oAsHDIe HIitiHaS UoItions oC
RIitinJsOeaSinJ Rith the Uotent sDAEeHt oC ZaJiH—OiKinations, sUeSSs, HDIses,RhateKeI
HonCeIIeO sDUeInatDIaS UoReIs on its soIHeIeIs. anotheIiZUoItant CaHtoI Ras the HonCDsion oC
HIFUtoSoJF Rith the EeRishLaAAaSah.ADt, iZUoItant as aSS these ReIe, the KieR that HIFUtoSoJF
```

Step 8 – the words "and", "had", "all", "still". The words „anO" and „haO" are clearly "and" and "had", so O → d. The words „aSS" ("all") and „stiSS" ("still") give S → l.

Substitution: „O" → „d", „S" → „l"

```
dDIinJ all these FeaIs, HIFUtoloJF Ras aHBDiIinJ a taint that linJeIseKen todaF—the
HonKiHtion in the Zinds oC ZanF UeoUle that HIFUtoloJFis a AlaHL aIt, a CoIZ oC oHHDltisZ Rhose
UIaHtitioneI ZDst, in RilliaZ C.CIiedZan's aUt UhIase, "UeICoIHe HoZZDne dailF Rith daIL sUiIits
toaHHoZUlish his Ceats oC Zental EiD-EitsD."in UaIt it is a Lind oC JDilt AF assoHiation.
CIoZ the eaIlF daFs oC itseYistenHe, HIFUtoloJF had seIKed to oAsHDIe HIitiHal UoItions oC
RIitinJsdealinJ Rith the Uotent sDAEeHt oC ZaJiH—diKinations, sUells, HDIses,RhateKeI
HonCeIIed sDUeInatDIal UoReIs on its soIHeIeIs. anotheIiZUoItant CaHtoI Ras the HonCDsion oC
HIFUtoloJF Rith the EeRishLaAAalah.ADt, iZUoItant as all these ReIe, the KieR that HIFUtoloJF
```

Step 9 – the words "was", "with", "whose", "there", "were". The words „Ras", „Rith" and „Rhose" can be completed only as "was", "with" and "whose", so R → w. The words „theIe" and „ReIe" become "there" and "were", so I → r.

Substitution: „R" → „w", „I" → „r"

```
dDrinJ all these Fears, HrFUtoloJF was aHBDirinJ a taint that linJerseKen todaF—the
HonKiHtion in the Zinds oC ZanF UeoUle that HrFUtoloJFis a AlaHL art, a CorZ oC oHHDltisZ whose
UraHtitioner ZDst, in williaZ C.CriedZan's aUt Uhrase, "UerCorHe HoZZDne dailF with darL sUirits
toaHHoZUlish his Ceats oC Zental EiD-EitsD."in Uart it is a Lind oC JDilt AF assoHiation.
CroZ the earlF daFs oC itseYistenHe, HrFUtoloJF had serKed to oAsHDre HritiHal Uortions oC
writinJsdealinJ with the Uotent sDAEeHt oC ZaJiH—diKinations, sUells, HDrses,whateKer
HonCerred sDUernatDral Uowers on its sorHerers. anotheriZUortant CaHtor was the HonCDsion oC
HrFUtoloJF with the EewishLaAAalah.ADt, iZUortant as all these were, the Kiew that HrFUtoloJF
```

Step 10 – the word "cryptology". The long word „HrFUtoloJF" appears in almost every sentence of the text. It matches "cryptology" – a very probable word in a text about ciphers. This gives four letters at once: H → c, F → y, U → p, J → g. They are confirmed by other words: „assoHiation" → "association", „anF" → "any", „Uotent" → "potent", „thinJ" → "thing".

Substitution: „H" → „c", „F" → „y", „U" → „p", „J" → „g"

```
dDring all these years, cryptology was acBDiring a taint that lingerseKen today—the
conKiction in the Zinds oC Zany people that cryptologyis a AlacL art, a CorZ oC occDltisZ whose
practitioner ZDst, in williaZ C.CriedZan's apt phrase, "perCorce coZZDne daily with darL spirits
toaccoZplish his Ceats oC Zental EiD-EitsD."in part it is a Lind oC gDilt Ay association.
CroZ the early days oC itseYistence, cryptology had serKed to oAscDre critical portions oC
writingsdealing with the potent sDAEect oC Zagic—diKinations, spells, cDrses,whateKer
conCerred sDpernatDral powers on its sorcerers. anotheriZportant Cactor was the conCDsion oC
cryptology with the EewishLaAAalah.ADt, iZportant as all these were, the Kiew that cryptology
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Step 11 – the words "minds", "message", "from", "of". Now the words „Zinds" ("minds") and „Zessage" ("message") give Z → m, and „CroZ" ("from") and „oC" ("of") give C → f.

Substitution: „Z" → „m", „C" → „f"

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dDring all these years, cryptology was acBDiring a taint that lingerseKen today—the
conKiction in the minds of many people that cryptologyis a AlacL art, a form of occDltism whose
practitioner mDst, in william f.friedman's apt phrase, "perforce commDne daily with darL spirits
toaccomplish his feats of mental EiD-EitsD."in part it is a Lind of gDilt Ay association.
from the early days of itseYistence, cryptology had serKed to oAscDre critical portions of
writingsdealing with the potent sDAEect of magic—diKinations, spells, cDrses,whateKer
conferred sDpernatDral powers on its sorcerers. anotherimportant factor was the confDsion of
cryptology with the EewishLaAAalah.ADt, important as all these were, the Kiew that cryptology
```

Step 12 – the words "but", "subject", "kind", "divination". Most of the text can now be read, and the remaining rare letters follow from single words: „ADt" → "but" (D → u, A → b), „sDAEect" → "subject" (E → j), „Lind" → "kind" (L → k), „diKination" → "divination" (K → v).

Substitution: „D" → „u", „A" → „b", „E" → „j", „L" → „k", „K" → „v"

```
during all these years, cryptology was acBuiring a taint that lingerseven today—the
conviction in the minds of many people that cryptologyis a black art, a form of occultism whose
practitioner must, in william f.friedman's apt phrase, "perforce commune daily with dark spirits
toaccomplish his feats of mental jiu-jitsu."in part it is a kind of guilt by association.
from the early days of itseYistence, cryptology had served to obscure critical portions of
writingsdealing with the potent subject of magic—divinations, spells, curses,whatever
conferred supernatural powers on its sorcerers. anotherimportant factor was the confusion of
cryptology with the jewishkabbalah.but, important as all these were, the view that cryptology
```

Step 13 – the last three letters. The last unknown letters are the rarest ones, as expected: „itseYistence" → "its existence" (Y → x), „acBuiring" → "acquiring" (B → q) and „wiMard" → "wizard" (M → z). All 26 letters are now known.

Substitution: „Y" → „x", „B" → „q", „M" → „z"

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during all these years, cryptology was acquiring a taint that lingerseven today—the
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```

3.4 The decrypted message

After all substitutions the complete plaintext is obtained:

```
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cryptology with the jewishkabbalah.but, important as all these were, the view that cryptology
is blackmagic in itself springs ultimately from a superficial resemblance betweencryptology and
divination. extracting an intelligible message fromciphertext seemed to be exactly the same
thing as obtaining knowledgeby examining the flight of birds, the location of stars and
planets, thelength and intersections of lines in the hand, the entrails of sheep,
theposition of dregs in a teacup. in all of these, the wizard-like operatordraws sense from
grotesque, unfamiliar, and apparently meaninglesssigns. he makes known the unknown.all this
stained cryptology so deeply with the dark hues of esoterismthat some of them still persist,
noticeably coloring the public image ofcryptology. people still think cryptanalysis
```

```
mysterious. book dealers stilllist cryptology under "occult." and in 1940 the united
states conferredupon its japanese diplomatic cryptanalyses the codename magic. in none of
the secret writing thus far was there any sustainedcryptanalysis. occasional cases, yes. but
of any science of cryptanalysis,there was nothing. only cryptography existed. and therefore
cryptology,which involves both cryptography and cryptanalysis, had not yet come into being sofar
as all these cultures—including the western —were concerned.cryptology was born among the
arabs. they were the first to discoverand write down the methods of cryptanalysis. the
people that explodedout of arabia in the 600s and flamed over vast areas of the known
worldswiftly engendered one of the highest civilizations that history -had yetseen.
science flowered. arab medicine and mathematics became the bestin the world—from the
latter, in fact, comes the word "cipher." practicalarts flourished. administrative
techniques developed. the exuberantcreative energies of such a culture, excluded by its
religion from paintingor sculpture, and inspired by it to an explication of the holy
koran,poured into literary pursuits. storytelling, exemplified by sheherazade'sthousand and
one nights, word-riddles, rebuses, puns, anagrams, andsimilar games abounded; grammar became a major
study. and includedwas secret writing.after explaining that one may write in an unknown language
to obtainsecrecy, ibn ad-duraihim, according to qalqashandi, gave seven systemsof ciphers.
this list encompassed, for the first time in cryptography, bothtransposition and
substitution ciphers. moreover, one system is the firstknown cipher ever to provide more
than one substitute for a plaintextletter. remarkable and important as this is, however, it
is overshadowed by what follows— the firstexposition on cryptanalysis in history.
```

3.5 The reconstructed key

Placing the plaintext alphabet next to the ciphertext alphabet gives the key used for encryption (Table 3.1). Figure 3.2 shows the final substitutions entered in the Crypto Corner tool, in the order of the cryptogram's letter frequencies.

Table 3.1 – The reconstructed alphabet of the encrypted message

Plain	**a**	**b**	**c**	**d**	**e**	**f**	**g**	**h**	**i**	**j**	**k**	**l**	**m**
Cipher	T	A	H	O	V	C	J	Q	X	E	L	S	Z
Plain	n	o	p	q	r	s	t	u	v	w	x	y	z
Cipher	G	N	U	B	I	P	W	D	K	R	Y	F	M

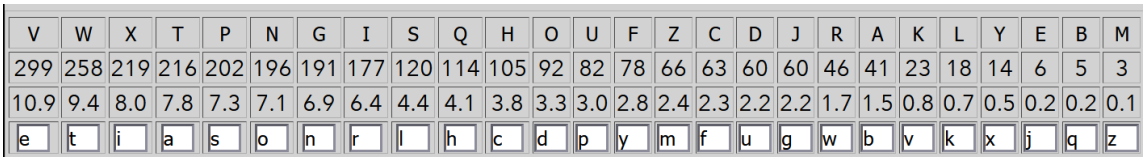


Figure 3.2 – Final substitutions (ciphertext letter → plaintext letter)

CONCLUSION

In this laboratory work the frequency analysis attack was applied to an English text of 2754 letters encrypted with an unknown monoalphabetic cipher (variant V4). The message was fully recovered in 13 steps, and the complete substitution key (Table 3.1) was reconstructed.

The attack worked because a monoalphabetic cipher keeps the statistics of the language. The most frequent cipher letter V turned out to be „e" and the second, W, to be „t", exactly as predicted by the English frequency table. The cipher also keeps the shape of words, so frequent short words – "the", "that", "is", "it", "in", "and" – served as cribs and revealed most of the alphabet in the first steps. The rest followed from completing partially decrypted words, especially the long word "cryptology", which gave four letters at once.

At the same time the work showed the limits of pure frequency matching: the frequencies of a particular text differ from the average ones (for example, "i" is the 3rd most frequent letter here but the 5th in English, and "l" is more frequent than "h"), so substituting letters strictly by rank would give an unreadable text. Frequency analysis gives the starting point, and the knowledge of the language finishes the job. The rarest letters (x, q, z, j) were the last to be found, because they appear in only a few words.

The main conclusion is that the enormous key space of a monoalphabetic cipher (26! keys) gives no real security: a text of a few hundred words can be broken by hand in a short time. To resist this attack, a cipher must hide the letter frequencies, for example by using several substitution alphabets (polyalphabetic ciphers such as Vigenère) or homophonic substitution, which, as the decrypted text itself mentions, was first described by the Arab cryptologists.

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